

EDS ASSIGNMENT 2

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Project Title: Comprehensive Data Analysis of Amazon E-commerce Performance

(Amazon Product Dataset)

Project Overview

The primary goal of this project is to extract actionable insights from the raw Amazon sales data. By cleaning, visualizing, and analyzing this data, we aim to uncover trends in product performance, pricing strategies, and customer sentiment, providing a data-driven perspective on how various products fare in an online marketplace.

Analysis of the 20 Problem Statements

Based on your assignment rubric, the 20 problem statements are designed to guide you through the full data science lifecycle. Below is a brief explanation of each:

Phase 1: Data Cleaning & Preprocessing

1. **Missing Value Imputation:** Handling incomplete data to ensure statistical accuracy.
2. **Duplicate Removal:** Cleaning the data to prevent skewed results from redundant records.
3. **Data Type Conversion:** Standardizing formats (e.g., converting text-based prices into numeric floats) for computation.
4. **Cleaning Text Data:** Standardizing categorical text for consistent grouping.
5. **Handling Outliers:** Detecting and addressing extreme values that could misrepresent performance.

Phase 2: Exploratory Data Analysis (EDA)

6. **Total Sales Calculation:** Deriving a core metric to measure overall financial performance.
7. **Best-Selling Products:** Identifying the "stars" of the product catalog.
8. **Revenue by Category:** Understanding which segments drive the most revenue.

9. **Customer Distribution:** Analyzing the reach of products based on user engagement.
10. **Geographical Analysis:** Mapping product performance across different regions or categories.

Phase 3: Temporal & Seasonal Trends

11. **Monthly Sales Trend:** Identifying when customers are most active.
12. **Peak Order Days:** Pinpointing which days of the week lead to higher volume.
13. **Hourly Order Analysis:** Understanding user behavior during different times of the day.
14. **Year-over-Year Growth:** Comparing performance milestones across different years.

Phase 4: Advanced Insights & Behavioral Analytics

15. **Payment Method Popularity:** Analyzing preferred transaction methods.
16. **Order Status Breakdown:** Assessing operational health (delivered vs. returned).
17. **Average Order Value (AOV):** Calculating the average spending behavior.
18. **Brand Performance:** Evaluating how specific brands compete in the market.
19. **Impact of Discounts:** Measuring the effectiveness of price-drop strategies on sales volume.
20. **Customer Retention:** Identifying repeat purchasers to gauge brand loyalty.

CODE AND OUTPUT

amazon-sale-dataset.ipynb ☆

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```
[5] import pandas as pd
import numpy as np

[6] df = pd.read_csv('amazon.csv')

[7] # 1. Missing Value Imputation (filling numeric with median, object with mode)
for col in df.select_dtypes(include=[np.number]).columns:
    df[col].fillna(df[col].median(), inplace=True)
for col in df.select_dtypes(include=['object']).columns:
    df[col].fillna(df[col].mode()[0], inplace=True)

... /tmp/ipykernel_5851/3847842887.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values al

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(

df[col].fillna(df[col].mode()[0], inplace=True)

[8] # 2. Duplicate Removal
df.drop_duplicates(inplace=True)

[9] # 3. Data Type Conversion (Cleaning prices first)
df['discounted_price'] = df['discounted_price'].replace({'$': '', ',': ''}, regex=True).astype(float)
df['actual_price'] = df['actual_price'].replace({'$': '', ',': ''}, regex=True).astype(float)
```

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```
[9] # 3. Data Type Conversion (Cleaning prices first)
df['discounted_price'] = df['discounted_price'].replace({'$': '', ',': ''}, regex=True).astype(float)
df['actual_price'] = df['actual_price'].replace({'$': '', ',': ''}, regex=True).astype(float)

[10] # 4. Cleaning Text Data
df['category'] = df['category'].str.lower().str.strip()

[11] # 5. Handling Outliers (e.g., in actual_price)
q_low = df['actual_price'].quantile(0.01)
q_hi = df['actual_price'].quantile(0.99)
df = df[(df['actual_price'] < q_hi) & (df['actual_price'] > q_low)]

[12] # 6. Total Sales Calculation (Assuming quantity is 1 if not provided)
df['TotalAmount'] = df['discounted_price']
# simplified for this dataset structure

[13] # 7. Best-Selling Products (by rating_count as a proxy for popularity)
best_selling = df.groupby('product_name')['rating_count'].sum().sort_values(ascending=False).head(10)

[14] # 8. Revenue by Category
cat_revenue = df.groupby('category')['discounted_price'].sum()

[15] # 9. Customer Distribution (Top 5 users)
top_users = df.groupby('user_name')['product_id'].count().sort_values(ascending=False).head(5)
```

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[15] 0s

9. Customer Distribution (Top 5 users)

top_users = df.groupby('user_name')['product_id'].count().sort_values(ascending=False).head(5)

[16] 0s

10. Geographical Analysis (Assuming review_id/user_id contains regional markers, here using category)

cat_dist = df['category'].value_counts()

[17] 0s

10.

Replace 'city' with your actual column name if it differs

if 'city' in df.columns:

geo_analysis = df['city'].value_counts()

print("Top Cities by Order Count:\n", geo_analysis.head())

[18] 0s

print(df.columns.tolist())

[product_id', 'product_name', 'category', 'discounted_price', 'actual_price', 'discount_percentage', 'rating', 'rating_count', 'about_produc

[19] 0s

13. Distribution of Ratings

rating_dist = df['rating'].value_counts().sort_index()

print("13. Rating Distribution:\n", rating_dist)

13. Rating Distribution:

rating

2 1

2.3 1

2.6 1

2.8 2

2.9 1

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1

3.0 3

3.1 4

3.2 2

3.3 16

3.4 10

3.5 26

3.6 34

3.7 42

3.8 84

3.9 123

4 51

4.0 128

4.1 238

4.2 224

4.3 218

4.4 120

4.5 74

4.6 17

4.7 5

4.8 3

5.0 3

| 1

Name: count, dtype: int64

[20] 0s

14. Average Discount per Category

Clean the discount percentage column first

df['discount_pct_num'] = df['discount_percentage'].astype(str).str.replace('%', '').astype(float)

avg_discount_by_cat = df.groupby('category')['discount_pct_num'].mean().sort_values(ascending=False)

print("\n14. Average Discount by Category:\n", avg_discount_by_cat.head())

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[20] 0s

avg_discount_by_cat = df.groupby('category')['discount_pct_num'].mean().sort_values(ascending=False)
print("\n14. Average Discount by Category:\n", avg_discount_by_cat.head())

...

14. Average Discount by Category:
category
computers&accessories|accessories|peripherals|cables&accessories|cableconnectionprotectors 90.0
electronics|headphones,earbuds&accessories|earpads 90.0
electronics|mobiles&accessories|mobileaccessories|décor|phonecharms 90.0
electronics|headphones,earbuds&accessories|adapters 88.0
computers&accessories|accessories|peripherals|keyboards,mice&inputdevices|keyboard&miceaccessories|dustcovers 87.5
Name: discount_pct_num, dtype: float64

[21] 0s

15. Find Top 5 Products with the Highest Discount Percentage
First, remove the '%' sign and convert to numeric
df['discount_pct_num'] = df['discount_percentage'].str.replace('%', '').astype(float)

Sort by the new numeric column in descending order
top_discounts = df[['product_name', 'discount_pct_num']].sort_values(by='discount_pct_num', ascending=False).head(5)

print("Top 5 Products with the Highest Discount:\n")
print(top_discounts)

Top 5 Products with the Highest Discount:

	product_name	discount_pct_num
695	rts [2 Pack] Mini USB C Type C Adapter Plug, T...	94.0
334	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91.0
372	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91.0
364	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91.0
368	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	91.0

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[22] 0s

16. Most Reviewed Brands
Extract brand (assuming it's the first word of the product_name)
df['brand'] = df['product_name'].str.split().str[0]
Clean rating_count
df['rating_count_num'] = df['rating_count'].str.replace(',', '').astype(float)
top_brands = df.groupby('brand')['rating_count_num'].sum().sort_values(ascending=False).head(5)
print("\n16. Top 5 Most Reviewed Brands:\n", top_brands)

...

16. Top 5 Most Reviewed Brands:
brand
boAt 4416568.0
Redmi 1878482.0
AmazonBasics 1808818.0
SanDisk 1426074.0
Amazon 1217337.0
Name: rating_count_num, dtype: float64

[23] 0s

17. Top Rated Product per Category
top Rated per_cat = df.loc[df.groupby('category')['rating'].idxmax()]['category', 'product_name', 'rating']
print("\n17. Top Rated Product per Category:\n", top Rated per_cat.head())

17. Top Rated Product per Category:

	category	
1157	car&motorbike caraccessories interioraccessori...	
695	computers&accessories accessories peripherals ...	
906	computers&accessories accessories peripherals ...	
748	computers&accessories accessories peripherals ...	

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695 computers&accessories|accessories&peripherals|...
906 computers&accessories|accessories&peripherals|...
748 computers&accessories|accessories&peripherals|...
936 computers&accessories|accessories&peripherals|...
```

	product_name	rating
1157	Reffair AX30 [MAX] Portable Air Purifier for C...	3.8
695	rts [2 Pack] Mini USB C Type C Adapter Plug, T...	4.3
906	Hp Wired On Ear Headphones With Mic With 3.5 M...	3.5
748	JBL Commercial CSLM20B Auxiliary Omnidirection...	3.9
936	ZEBRONICS Zeb-Warrior II 10 watts 2.0 Multimed...	4.1

```
[24] ✓ 0s
# 18. Brand Performance (Extracting brand from product_name)
df['brand'] = df['product_name'].str.split().str[0]
brand_perf = df.groupby('brand')['discounted_price'].sum()
```

```
[25] ✓ 0s
# 19. Impact of Discounts
df['discount_pct_num'] = df['discount_percentage'].replace({'%': ''}, regex=True).astype(float)
discount_corr = df['discount_pct_num'].corr(df['discounted_price'])
```

```
[26] ✓ 0s
# 20. Customer Retention
repeat_customers = df['user_id'].value_counts()
repeat_customers = repeat_customers[repeat_customers > 1].count()
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